Joseph Pongonthara

647-390-1768 | jpongont@uwaterloo.ca | linkedin.com/in/joseph-pongonthara | https://josephpongonthara.github.io/site/

Professional Summary

Mechatronics Engineering graduate with a strong foundation in Object-Oriented Design, problem-solving, and statistical analytics. Eager to apply my diverse problem solving skills to complex problems at Dominion Dynamics.

Education

University of Waterloo — Waterloo, Canada
Bachelor of Applied Science in Mechatronics Engineering — *Graduated 2025*

St. Michael's Choir School — Toronto, Canada
Valedictorian, Student Council President — *Graduated 2019*

Technical Skills

- **Languages:** Python, Javascript, TypeScript, C++, Embedded C, SQL, MATLAB, C#
- **Data & Machine Learning:** Hadoop Cloud Infrastructure, Data Wrangling, Data Preprocessing, Data Visualization, (Familiar with: Pandas, NumPy, Scikit-learn)
- **Software & MLOps:** Git, JIRA, REST APIs, Selenium, CI/CD Pipelines
- **Backend & Cloud:** Azure, PostgreSQL, Supabase, AWS (familiar), Cosmos DB
- **Core Competencies:** Algorithm Design, Data Structures, Automation, Quantitative Analysis, Distributed Systems, Database Management

Professional Employment History

Senior Analyst - Credit Risk Modeling — BMO — February 2026 - Present

- Conduct quantitative analysis of IFRS9 and CECL credit-risk model outputs, obtaining ECL, EAD, PD and LGD across portfolios, scenarios, horizons, borrower segments, etc, for audit and regulatory agencies.
- Leverage PySpark, Spark SQL, SAS, Netezza and Hadoop/Cloudera to transform, aggregate and integrate large-scale credit-risk datasets for various pieces of analysis.
- Perform root-cause analysis, identifying discrepancies across source data, mappings, joins and reporting outputs for data reconciliation and model integrity purposes, ensuring completeness and accuracy.
- Working alongside a team of analysts and machine learning engineers to develop a MCP server between github copilot and hadoop cloudera resources to allow for automation on recurring risk analytics to improve the repeatability and efficiency of portfolio-level ECL and model-metric analysis.

AI Prototype Engineer — Tenuto Labs — May 2025 - December 2025

- Worked as part of a small team throughout the software development life cycle; conducted technical discovery sessions and requirements gathering with stakeholders, including C-level executives, iteratively translating user feedback into clearly defined MVP features and functional software solutions.
- Designed AI-powered data transformation prototypes to Azure-based infrastructure, implementing secure API authentication, telemetry logging, and automated test coverage for reliability at scale.
- Designed and documented a scalable backend architecture using PostgreSQL (via Supabase) and Azure, ensuring secure, real-time data delivery for AI-powered solutions.
- Integrated continuous stakeholder feedback into feature design, applying agile best practices to deliver a prototype ready for MVP production in 10 weeks.

Engineering Intern — Callidus Engineering, Quasar Consulting — Jan - Apr, Sep - Dec 2022, Jan - Aug 2024

- Interviewed engineering and design teams to understand workflow bottlenecks, then developed Python automation tools that reduced their manual review time by 30%.
- Standardized CAD workflows and implemented version control for drawing revisions via Git-based documentation system.
- Delivered high-volume AutoCAD and Revit drawing packages for large-scale residential and commercial builds, often under aggressive timelines.

QA Automation Intern — i4i — January - April, September - December 2020;

- Developed and maintained Python/Selenium automated test suites for pharmaceutical regulatory documentation tools.
- Created test plans & automated regression suites, integrated with JIRA and internal dev pipelines.